

Degree-Adjusted Lesion Network Mapping (DA-LNM): A Compact, Volumetric Pipeline Built on a Low-Rank Normative Connectome

Methods

Cognitive Neuroscience and Clinical Phenomenology Laboratory

1. Overview and Rationale

Lesion network mapping (LNM) localizes the symptoms produced by focal brain lesions not to the lesion itself but to the distributed brain network to which the lesion is functionally connected. Rather than testing whether damage to a single voxel predicts a deficit, LNM embeds the lesion into a normative functional connectome derived from healthy individuals and asks which brain regions are coupled to the lesioned territory. The canonical procedure comprises three steps: (i) a lesion is delineated on a patient scan and mapped to a standard template; (ii) the lesion volume is used as a seed for resting-state functional connectivity computed within a normative connectome; and (iii) the resulting connectivity maps are analysed statistically to yield a lesion network map (Boes et al., 2015; Fox, 2018; Joutsa et al., 2022).

A recurring concern with standard LNM is that thresholded network maps can be dominated by the global connectivity structure of the normative connectome rather than by lesion-specific topography. Recent formal analyses show that averaging-and-thresholding LNM approximates a linear operation in which the output collapses toward the row-summation vector—the *weighted degree*—of the connectome matrix, so that maps for unrelated syndromes converge on the same high-degree hubs (van den Heuvel et al., 2026). This does not invalidate LNM, but it means that specificity must be established explicitly, for example by removing the degree contribution or by testing maps against a connectome-aware null (Dimensionality-reduction specificity work, 2026).

Design goals. The pipeline described here was built to meet four practical requirements simultaneously: (1) it must accept an ordinary volumetric lesion mask in MNI space and return a voxelwise network map on the same grid; (2) it must preserve across-subject variance so that a genuine one-sample *t*-statistic can be formed and thresholded as in classical LNM; (3) it must be light enough to embed in the back end of an application, avoiding the tens of gigabytes of a full voxelwise connectome; and (4) it must adjust for the normative-connectome degree so that the reported network is not merely hub structure. The result is a two-stage system: an offline *preparation* stage that compiles a compact connectome bundle, and a runtime *back end* that computes a degree-adjusted network map per lesion.

2. Normative Connectome Substrate

The normative substrate is the Human Connectome Project (HCP) S1200 release, specifically its Parcellation + Timeseries + Netmats (PTN) derivative built from the 1003 subjects with four complete resting-state runs (Van Essen et al., 2013; Smith et al., 2013). Rather than represent connectivity as a dense voxel-by-voxel matrix, the PTN release provides a group functional decomposition obtained by spatial group independent component analysis (group-ICA; MELODIC) at several dimensionalities, together with subject-specific node timeseries estimated by multiple spatial regression against the group maps—the first stage of dual regression (Beckmann & Smith, 2004; Filippini et al., 2009). All data were surface-aligned with MSMAll (Robinson et al., 2014) and minimally preprocessed with the HCP pipelines (Glasser et al., 2013).

Two PTN products are used. The volumetric group-ICA spatial maps, distributed as `melodic_IC_sum.nii.gz` in MNI152 2 mm space (the components dual-regressed into subjects' 3-D data and averaged), provide a

spatial basis in the same volumetric frame as a lesion mask; this avoids any surface/grayordinate conversion of the lesion. The per-subject node timeseries provide the *subject-level temporal information* required to retain across-subject variance. The chosen dimensionality D (typically 100 or 200) sets the rank of the connectome model; larger D resolves finer spatial detail at the cost of a larger bundle.

Conceptually the pipeline treats connectivity as a rank- D model. If A denotes the volumetric spatial maps (voxels $\times$ components) and G the group temporal covariance among components, the implied voxelwise connectome is the low-rank product $\hat{C} = A G A^T$. Seed-to-voxel connectivity, degree, and their statistics are all derived from this factorized form without ever materializing $\hat{C}$.

3. Preparation Stage (prepare_connectome.py)

The preparation stage is executed once. It reads the PTN products for one dimensionality and writes a single compressed bundle (.npz) that the back end loads thereafter. The raw HCP archives are not touched again.

3.1 Volumetric spatial basis

The 4-D file `melodic_IC_sum.nii.gz` is loaded as an array of shape (X, Y, Z, D) and reshaped to a matrix of shape $(Vall \times D)$. A brain mask is defined as the set of voxels with any non-zero component loading, and the maps are restricted to those voxels, giving the spatial-basis matrix

$$A \in \mathbb{R}^{(V \times D)}, \quad A[v, c] = \text{loading of component } c \text{ at voxel } v.$$

The mask, the image affine, and the grid dimensions are retained so that outputs can be written back to the identical space.

3.2 Per-subject temporal covariance

For each connectome subject i ($i = 1 \dots N$), the node-timeseries matrix T_i of shape $(T_i \times D)$ is loaded and column-demeaned, and its temporal covariance across components is computed:

$$\Sigma_i = (T_i^T T_i) / (T_i - 1) \in \mathbb{R}^{(D \times D)}.$$

Storing the $D \times D$ covariance rather than the full timeseries is the key to the compact footprint: the lesion-specific seed connectivity for any lesion can be reconstructed exactly from Σ_i alone (Section 4.3), so the many long timeseries files are never needed at runtime. The per-subject covariances are stacked into an array of shape $(N \times D \times D)$.

3.3 Group covariance

The group temporal covariance, used later to derive the connectome degree, is the mean across subjects:

$$G = (1/N) \Sigma_i \Sigma_i \in \mathbb{R}^{(D \times D)}.$$

3.4 Bundle contents

The stage writes a single bundle containing the arrays below (approximate on-disk sizes: ~100 MB for $D = 100$, ~300 MB for $D = 200$):

Array	Shape	Description
	$V \times D$	Volumetric group-ICA spatial maps at in-brain voxels (the rank- D basis).
	$X \times Y \times Z$	Boolean brain mask defining the V voxels and the output frame.

Array	Shape	Description
	4×4	Image-to-world affine of the MNI152 2 mm grid.
	3	Volume dimensions (X, Y, Z).
	$N \times D \times D$	Per-subject temporal covariance stack (Σ_i).
	$D \times D$	Group temporal covariance (for the degree map).
	scalar	The ICA dimensionality D.

4. Runtime Back End (lnm_backend.py)

The back end loads the bundle once and exposes a single operation: given a lesion mask, return a degree-adjusted voxelwise t-map plus a network summary. All steps are closed-form linear-algebra operations on the bundle arrays; no per-request iteration over subjects or voxels is required.

4.1 Lesion ingestion and spatial normalization

The lesion image is loaded and resampled to the bundle grid by nearest-neighbour interpolation (preserving the binary mask), then binarized at 0.5 and intersected with the brain mask to yield an index set L of in-brain lesion voxels. Because resampling is defined through the images' affines in world coordinates, lesions delineated at any MNI resolution (e.g., 1 mm) are mapped correctly onto the 2 mm connectome grid. A lesion that does not overlap the brain mask after resampling raises an explicit error, guarding against space or orientation mismatches.

4.2 Seed definition

The lesion is expressed in component space as a seed loading vector—the mean spatial-basis profile over the lesion voxels:

$$w = (1/|L|) \sum_{v \in L} A[v, :] \in \mathbb{R}^D.$$

Equivalently, the seed timeseries for subject i is the component-weighted combination $s_i(t) = \sum_c w_c T_i(t, c) = T_i w$. A least-squares spatial regression of the lesion onto A is a drop-in alternative; the mean-overlap definition is used by default for robustness with small lesions.

4.3 Per-subject seed-to-node connectivity

For each subject the correlation between the seed timeseries and every component is obtained analytically from that subject's covariance, using $\text{cov}(s_i, c) = w^T \Sigma_i e_c$, $\text{var}(s_i) = w^T \Sigma_i w$, and $\text{var}(\text{node } c) = \Sigma_i(c, c)$:

$$r_i(c) = (w^T \Sigma_i e_c) / \sqrt{(w^T \Sigma_i w) \cdot \Sigma_i(c, c)}.$$

This reproduces the correlation that would be obtained from the full timeseries, but requires only Σ_i . The result is a subject-by-component array of seed-to-node correlations.

4.4 Variance-stabilizing transform

Correlations are Fisher-transformed to approximate normality prior to group statistics:

$$z_i(c) = \text{arctanh}(r_i(c)).$$

4.5 Low-rank voxelwise one-sample t-map

Each subject's seed-to-voxel connectivity is the low-rank reconstruction $m_i = A z_i$. Rather than build the $N \times V$ stack of subject maps, the across-subject mean and variance at every voxel are obtained in closed form from the component-space mean $\bar{z}$ and covariance $C_z = \text{cov}(z_i)$:

$$\text{mean}(v) = (A \bar{z})(v) \quad \text{Var}(v) = (A C_z A^T)(v, v) = \sum_c (A C_z)[v, c] \cdot A[v, c].$$

The voxelwise one-sample t-statistic across the N connectome subjects is then

$$t(v) = \text{mean}(v) / \sqrt{\text{Var}(v) / N}.$$

This is the exact analogue of the classical LNM group t-test over per-subject connectivity maps, computed without ever forming a large subject-by-voxel matrix. Positive values denote voxels positively coupled to the lesion network; negative values denote anticorrelated regions.

4.6 Connectome degree

The weighted degree of a voxel is its total connectivity to the rest of the brain, i.e., the row sum of the implied connectome $\hat{C} = A G A^T$. Exploiting the factorization, the full degree map is obtained with two small matrix products and never requires $\hat{C}$:

$$g = A^T 1 \text{ (component column sums)} \quad d = A (G g) \in \mathbb{R}^V.$$

d is lesion-independent and is precomputed once when the bundle is loaded.

4.7 Degree adjustment

To remove the trivial degree contribution, the t -map is residualized on the degree map by ordinary least squares. With a design $X = [1, \tilde{d}]$ ($\tilde{d}$ standardized) and coefficients $\hat{\beta} = (X^T X)^{-1} X^T t$, the degree-adjusted map is

$$t^*(v) = t(v) - (X \hat{\beta})(v).$$

The pipeline reports the spatial correlation between the map and the degree vector before and after adjustment, $p_{\text{before}} = \text{corr}(t, d)$ and $p_{\text{after}} = \text{corr}(t^*, d)$. A large p_{before} reduced to near zero confirms that hub structure has been removed and that the retained topography is lesion-specific. The degree-adjusted map t^* is the primary output; the unadjusted map t is optionally saved for comparison.

4.8 Thresholding, signed mask, and atlas labelling

At a chosen threshold τ (default $|t| \geq 7$), the positive network $\{v : t^*(v) \geq \tau\}$ and negative network $\{v : t^*(v) \leq -\tau\}$ are extracted and written as a single signed mask (+1 coupled, -1 anticorrelated, 0 elsewhere). Each tail is labelled against two atlases resampled to the output grid: the Harvard-Oxford cortical and subcortical maximum-probability atlases for anatomical attribution (Desikan et al., 2006; Makris et al., 2006), and the Yeo seven-network parcellation for functional-network attribution (Yeo et al., 2011). For every region the number of network voxels, its percentage of the tail, and the mean t are tabulated and the leading regions reported, giving an immediate, interpretable summary alongside the volumetric map.

5. Statistical and Theoretical Considerations

Rank-D approximation. Seed-to-voxel connectivity here is a rank-D reconstruction through the group-ICA basis; spatial detail is therefore bounded by D . This is the same low-dimensional manifold view that motivates recent specificity analyses of LNM (2026), and it is what makes the model compact. $D = 200$ is recommended for final maps, with $D = 100$ for rapid iteration.

Relation to the degree critique. Because the model is explicitly linear (maps are A applied to seed-to-node statistics), it is transparent about the very property van den Heuvel et al. (2026) identify: the unadjusted map's alignment to degree is measurable and, here, is measured and removed. The degree adjustment is a linear (first-order) correction and does not remove non-linear dependence on degree, which is a second-order concern for most applications.

Averaged versus per-subject statistics. Collapsing a connectome to a single averaged matrix yields only a correlation/ z map and forfeits the across-subject variance needed for a t -test. By retaining per-subject covariances, this pipeline recovers a legitimate one-sample t -map at a footprint comparable to an averaged connectome, so the familiar t -thresholding of classical LNM remains available.

6. Variables, Dependencies, and Usage

6.1 Software dependencies

Package	Source	Purpose
Python $\geq$ 3.9	python.org	Runtime interpreter (tested on 3.10).
numpy	PyPI	Array algebra: covariance, einsum, least-squares.
nibabel	PyPI	Reading/writing NIfTI images and affines.
nilearn	PyPI	Lesion resampling and atlas fetching/labelling.

Install with: `pip install numpy nibabel nilearn`. The first back-end run with atlas labelling downloads the Harvard-Oxford and Yeo atlases once (cached under `~/nilearn_data`); pass `--no-atlas` for a fully offline run.

6.2 Data dependencies

Input	Description
<code>melodic_IC_sum.nii.gz</code>	Volumetric group-ICA maps (MNI152 2 mm), one per chosen dimensionality, from the HCP S1200 PTN groupICA archive.
<code>node_timeseries/*.txt</code>	Per-subject node timeseries (one file per subject) for the chosen dimensionality, from the HCP S1200 PTN NodeTimeseries archive.
<code>lesion.nii(.gz)</code>	Binary lesion mask in MNI space (any MNI resolution; resampled internally).

Access terms. HCP data are distributed under the WU-Minn HCP Open Access Data Use Terms, which restrict redistribution to users who have themselves registered and accepted the terms. The derived bundle should be treated accordingly; embedding HCP-derived matrices in distributed software should be checked against the current terms. The GSP1000 normative connectome is an alternative substrate where those terms are impractical.

6.3 Key parameters and variables

Symbol / flag	Meaning	Default / where set
<code>D / --dim</code>	ICA dimensionality = rank of the connectome model.	100 (prepare stage)
<code>N</code>	Number of connectome subjects.	~1003 (from PTN)
<code>V</code>	Number of in-brain voxels in the mask.	derived from maps
<code>w</code>	Seed loading vector (lesion in component space).	computed per lesion
<code>τ / --threshold</code>	$ t $ threshold for network / mask / labelling.	7.0
<code>--top</code>	Number of atlas regions listed per tail.	10
<code>SAVE_RAW</code>	Also write the unadjusted t-map.	True
<code>DEFAULT_LESION</code>	Lesion path; all output names derive from it.	edit at top of back end
<code>DEFAULT_BUNDLE</code>	Path to the prepared .npz bundle.	edit at top of back end

6.4 Usage — preparation (run once per dimensionality)

```
# edit DEFAULT_ROOT / DEFAULT_DIM at the top, then press Run,
```

```
# or pass them on the command line:
python prepare_connectome.py \
    --root "...\\HCP1200_Parcellation_Timeseries_Netmats\\HCP_PTN1200" \
    --dim 100
```

-> writes lnm_bundle_d100.npz into the root folder

Progress prints an [ica] line, a subject counter climbing to ~1003, then [done] with the bundle path and size. Repeat with --dim 200 to build the higher-resolution bundle.

6.5 Usage — back end (per lesion)

```
# edit DEFAULT_LESION (and DEFAULT_BUNDLE) at the top, then press Run,
# or pass on the command line:
python lnm_backend.py \
    --lesion "...\\Therasa_294069P.nii" \
    --bundle "...\\lnm_bundle_d100.npz" \
    --threshold 7
```

optional flags:

```
# --out / --mask-out / --raw-out  override auto-named outputs
# --raw-out ""                    do not save the unadjusted map
# --mean                          save mean Fisher-z instead of t
# --no-atlas                       skip atlas download/labelling
```

Output file names are derived automatically from the lesion filename; only the lesion path need change between runs. Programmatic use mirrors the CLI:

```
from lnm_backend import Connectome, summarize_network
c = Connectome(bundle_path) # load once at startup
res = c.compute(lesion_path, out_path='tmap.nii.gz')
summ = summarize_network(res['img'], threshold=7, verbose=False)
```

6.6 Outputs

File / value	Description
<lesion>_lnm_tmap.nii.gz	Degree-adjusted voxelwise t-map (primary output), MNI152 2 mm.
<lesion>_lnm_tmap_RAW.nii.gz	Unadjusted t-map (optional), for before/after comparison.
<lesion>_network_mask.nii.gz	Signed thresholded mask: +1 coupled, -1 anticorrelated.
degree r before -> after	Spatial correlation with degree, quantifying the adjustment.
Console region tables	Harvard-Oxford (anatomy) and Yeo-7 (network) summaries per tail.

7. Limitations

Three limitations follow from the design. First, connectivity is a rank-D approximation, so structure finer than the ICA basis is not recoverable; reporting the dimensionality used is essential. Second, the seed is defined by mean lesion overlap with the components, which is robust but coarse for very small lesions, where a least-squares seed may be preferable. Third, the normative connectome is the HCP young-adult cohort; applications in older or clinical populations inherit any age- or cohort-related differences in functional architecture, and substrate choice (HCP versus GSP1000, global-signal-regression versus aCompCor preprocessing) can influence anticorrelated components of the map.

8. Data and Code Availability

The normative data are available from the HCP S1200 release via ConnectomeDB under the WU-Minn HCP Open Access Data Use Terms. The pipeline comprises two Python modules, `prepare_connectome.py` (bundle preparation) and `lnm_backend.py` (runtime mapping and summary). Any publication using HCP data should include the acknowledgement required by the HCP consortium.

9. References

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Note: the 2026 methodological references are recent works on the degree/specificity properties of LNM; readers should verify the final citation details and venue against the current literature.